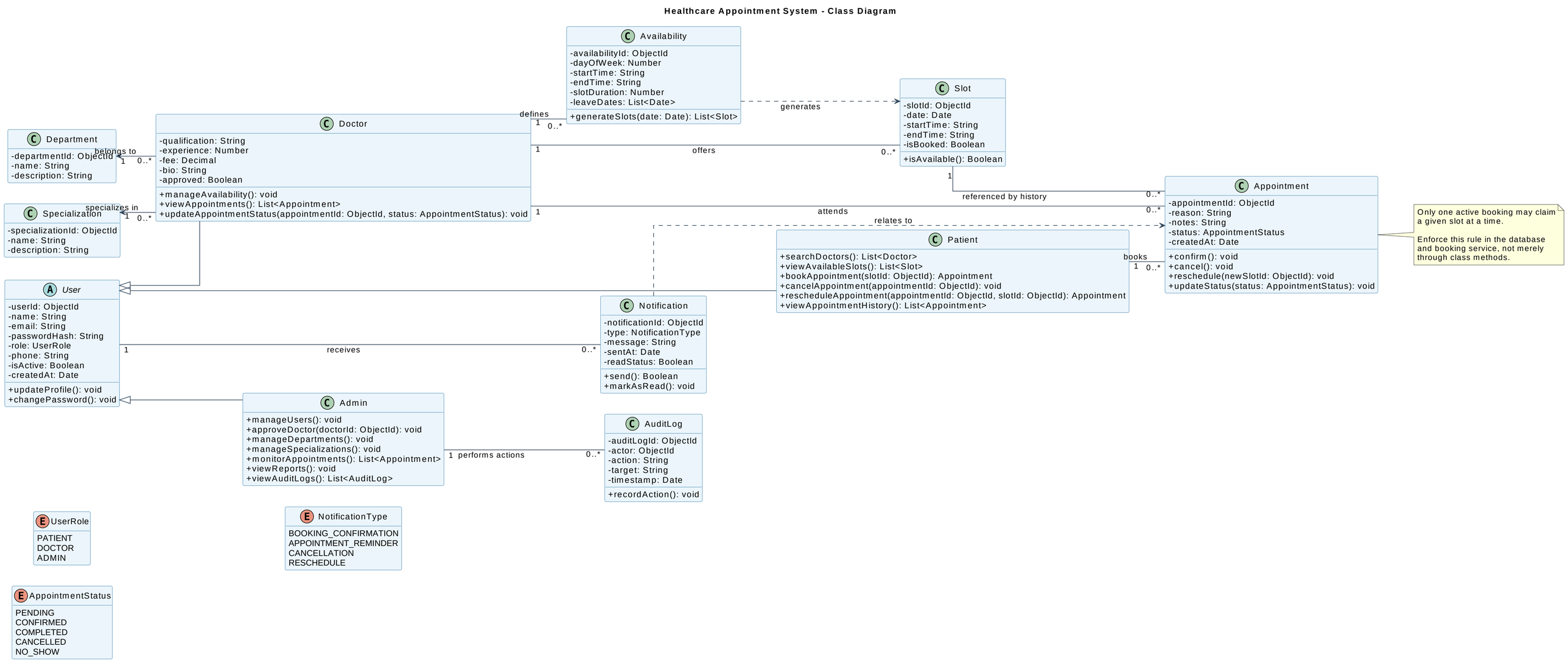
Healthcare Appointment System - Class Diagram
Availability
-availabilityId: ObjectId
-dayOfWeek: Number
-startTime: String
-endTime: String
-slotDuration: Number
-leaveDates: List<Date>
+generateSlots(date: Date): List<Slot>
generates
Slot
-slotId: ObjectId
-date: Date
-startTime: String
-endTime: String
-isBooked: Boolean
+isAvailable(): Boolean
defines
1
0..*
Department
-departmentId: ObjectId
-name: String
-description: String
belongs to
1
0..*
Doctor
-qualification: String
-experience: Number
-fee: Decimal
-bio: String
-approved: Boolean
+manageAvailability(): void
+viewAppointments(): List<Appointment>
+updateAppointmentStatus(appointmentId: ObjectId, status: AppointmentStatus): void
1
offers
0..*
1
referenced by history
0..*
Appointment
-appointmentId: ObjectId
-reason: String
-notes: String
-status: AppointmentStatus
-createdAt: Date
+confirm(): void
+cancel(): void
+reschedule(newSlotId: ObjectId): void
+updateStatus(status: AppointmentStatus): void
Only one active booking may claim a given slot at a time.
Enforce this rule in the database and booking service, not merely through class methods.
Specialization
-specializationId: ObjectId
-name: String
-description: String
specializes in
1
0..*
1
attends
0..*
relates to
Patient
+searchDoctors(): List<Doctor>
+viewAvailableSlots(): List<Slot>
+bookAppointment(slotId: ObjectId): Appointment
+cancelAppointment(appointmentId: ObjectId): void
+rescheduleAppointment(appointmentId: ObjectId, slotId: ObjectId): Appointment
+viewAppointmentHistory(): List<Appointment>
books
1
0..*
User
-userId: ObjectId
-name: String
-email: String
-passwordHash: String
-role: UserRole
-phone: String
-isActive: Boolean
-createdAt: Date
+updateProfile(): void
+changePassword(): void
Notification
-notificationId: ObjectId
-type: NotificationType
-message: String
-sentAt: Date
-readStatus: Boolean
+send(): Boolean
+markAsRead(): void
1
receives
0..*
Admin
+manageUsers(): void
+approveDoctor(doctorId: ObjectId): void
+manageDepartments(): void
+manageSpecializations(): void
+monitorAppointments(): List<Appointment>
+viewReports(): void
+viewAuditLogs(): List<AuditLog>
1
performs actions
0..*
AuditLog
-auditLogId: ObjectId
-actor: ObjectId
-action: String
-target: String
-timestamp: Date
+recordAction(): void
UserRole
PATIENT
DOCTOR
ADMIN
NotificationType
BOOKING_CONFIRMATION
APPOINTMENT_REMINDER
CANCELLATION
RESCHEDULE
AppointmentStatus
PENDING
CONFIRMED
COMPLETED
CANCELLED
NO_SHOW